

Set B. Macroeconomics Case Study Practice — Short-Answer Questions

Instructions to students. This booklet contains four case studies on contemporary macroeconomic problems: Singapore’s productivity slowdown and the wage–inflation transmission; the United Kingdom’s post-Brexit current account deficit and trade-cost adjustment; Indonesia’s FDI-driven nickel-processing strategy and exchange-rate management; and Vietnam’s high-growth, high-inflation development trajectory. Each case study contains an extract and a data table, followed by short-answer questions ranging from 2 to 6 marks.

Use the mark allocation as a guide to depth: 2-mark questions require a precise definition or identification with reference to the data; 4-mark questions require a developed two-step explanation; 6-mark questions require a fully developed analytical response supported by an AD/AS diagram or relevant data. Write your answers in the lined spaces provided.

Section Case Study 1. Singapore — Productivity Slowdown and Wage-Driven Inflation

Table 1.1: Selected Singapore Labour Market and Productivity Indicators, 2018–2023

Indicator	2018	2019	2020	2021	2022	2023
Nominal Wage Growth, total (%)	4.1	3.4	−1.5	3.9	6.5	5.2
Real Wage Growth, total (%)	3.7	2.8	−1.3	1.6	0.4	0.4
Labour Productivity Growth, VA per hour (%)	3.3	0.4	−2.1	5.4	0.4	−1.9
Unit Labour Cost, growth (%)	0.7	3.0	0.6	−1.4	6.1	7.1
Foreign Workforce Share (% of total)	38.4	38.7	36.5	33.7	33.0	33.9
MAS Core Inflation (%)	1.7	1.0	−0.2	0.9	4.1	4.2
Resident Unemployment Rate (%)	2.9	3.1	4.1	3.5	2.9	2.8

Source: Adapted from Ministry of Manpower Singapore Yearbook of Manpower Statistics 2023, MAS Macroeconomic Review, and Department of Statistics Singapore.

Extract 1.1: Wage gains squeeze Singapore firms as productivity stalls

Singapore experienced a dual labour market dynamic in 2022 and 2023 that has alarmed economists and policymakers in equal measure. Nominal wage growth rose to 6.5 per cent in 2022 and 5.2 per cent in 2023, well above the long-run average of about 3.5 per cent. The Manpower Ministry attributed the surge to a tight labour market, with the resident unemployment rate falling to 2.8 per cent and severe shortages reported in healthcare, construction, food services, and information technology.

However, labour productivity growth, measured as real value-added per hour worked, was just 0.4 per cent in 2022 and contracted by 1.9 per cent in 2023. The result was that unit labour costs — nominal wages divided by labour productivity — surged by 6.1 per cent and 7.1 per cent in those two years, the steepest sustained increase in Singapore’s recorded history. “Firms cannot pass on rising costs indefinitely, but the alternative is to absorb them and erode competitiveness or pass them on and stoke inflation,” a senior economist at MAS told the Business Times.

The productivity stagnation has multiple causes. The 2020–2021 reduction in foreign workforce share from 38.7 to 33.7 per cent — reflecting Covid-era border restrictions and subsequent quota tightening — forced firms to substitute more expensive resident labour for previously cheaper foreign hires without offsetting productivity gains. The Productivity and Innovation Credit scheme expired in 2018 and its successor, the Productivity Solutions Grant, has reached only a modest share of small and medium-sized enterprises. Meanwhile, Singapore’s services sector, which now accounts for over 70 per cent of GDP, has structurally lower measured productivity growth than the manufacturing sectors that drove earlier decades of catch-up growth.

The MAS Core Inflation reading of 4.2 per cent in 2023 reflects this wage–productivity disjuncture. “We are no longer dealing with a one-off imported cost shock,” MAS warned in its April 2024 Macroeconomic Review. “Domestic services inflation, which is tightly linked to wage costs, has become the principal driver of underlying inflation.”

Source: Adapted from The Business Times, 19 February 2024 and MAS Macroeconomic Review, April 2024.

Question 1**[2 marks]**

With reference to Table 1.1, calculate the change in Singapore’s real wage growth between 2019 and 2022. Show your working.

Question 2**[2 marks]**

Define “unit labour cost” as used in Extract 1.1.

Question 3**[4 marks]**

Using evidence from Extract 1.1 and Table 1.1, explain how the contraction in Singapore’s foreign workforce share contributed to the rise in unit labour costs after 2020.

Question 4**[4 marks]**

Explain how rising unit labour costs of 7.1 per cent in 2023 (Table 1.1) could affect the international price competitiveness of Singapore's exports.

Question 5**[6 marks]**

With the aid of an AD/AS diagram, explain how a sustained rise in unit labour costs, as described in Extract 1.1, would generate domestic services inflation in Singapore.

Question 6**[6 marks]**

Explain two supply-side policies the Singapore government could adopt to address the productivity stagnation described in Extract 1.1.

Section Case Study 2. United Kingdom — Post-Brexit Trade Costs and the Current Account Deficit

Table 2.1: Selected United Kingdom External Sector Indicators, 2016–2023

Indicator	2016	2017	2018	2019	2020	2021	2022	2023
Current Account Balance (% GDP)	−5.2	−3.8	−3.7	−2.7	−2.5	−0.5	−3.0	−3.5
Goods Trade Balance (% GDP)	−6.5	−5.8	−5.9	−5.8	−5.0	−5.7	−8.0	−7.5
Services Trade Balance (% GDP)	5.0	4.9	4.7	5.0	5.0	5.1	5.2	5.4
Pound Sterling Effective Index (2010=100)	89	85	84	84	84	88	84	86
UK Goods Exports to EU (£ billion)	137	153	162	167	138	159	192	188
UK Real GDP Growth (%)	1.9	2.4	1.7	1.6	−10.4	8.7	4.3	0.1

Source: Adapted from Office for National Statistics UK Balance of Payments Pink Book 2023 and Bank of England.

Extract 2.1: Brexit, services exports, and the persistent goods deficit

The United Kingdom’s current account has been in deficit every year since 1984, but the post-Brexit pattern is distinctive: the deficit has shifted decisively from a deficit-with-large-services-offset model toward a deepening goods deficit that services exports can no longer fully offset.

The UK’s withdrawal from the European Union single market in January 2021 introduced non-tariff trade barriers (NTBs) including customs declarations, sanitary and phytosanitary checks, and rules-of-origin requirements. The Office for Budget Responsibility estimates that these NTBs raised the effective trade cost between the UK and the EU by approximately 14 percentage points, equivalent to a substantial tariff that did not previously exist. UK goods exports to the EU fell by 18 per cent in 2021 before partially recovering as firms adjusted. Notably, small and medium-sized exporters were disproportionately affected, with many ceasing EU exports entirely rather than absorbing the compliance cost.

The services sector has performed strikingly well in contrast. The UK’s services trade balance widened to 5.4 per cent of GDP by 2023, supported by financial services, professional consulting, and education. However, this resilience reflects two factors that may not persist. First, services trade is less impeded by the EU–UK Trade and Cooperation Agreement, which provides relatively limited market access for services. Second, the post-Covid recovery in international tourism flows favoured the UK disproportionately.

A separate concern is the deterioration in the primary income balance, which moved from a small surplus to a deficit of approximately 1 per cent of GDP. This reflects falling investment income on UK foreign assets and rising income outflows to foreign owners of UK assets, partly attributable to the post-2016 fall in the pound that reduced the relative value of UK assets in foreign-currency terms. The Bank of England warned in its January 2024 Financial Stability Report that the combination of a widened goods deficit, weakening primary income, and elevated public-sector borrowing leaves the UK “unusually reliant on continued foreign portfolio inflows.”

Source: Adapted from Office for Budget Responsibility Economic and Fiscal Outlook, November 2023 and Bank of England Financial Stability Report, January 2024.

Question 1

[2 marks]

With reference to Table 2.1, describe what happened to the UK's goods trade balance between 2019 and 2023.

Question 2**[2 marks]**

Define “non-tariff trade barriers” as used in Extract 2.1.

Question 3**[4 marks]**

Using evidence from Extract 2.1, explain how the post-Brexit non-tariff trade barriers contributed to the widening of the UK's goods trade deficit.

Question 4**[4 marks]**

With reference to Table 2.1, explain why the UK's services trade surplus has been insufficient to offset the goods trade deficit since 2021.

Question 5**[6 marks]**

With the aid of a relevant diagram, explain how a depreciation of the pound sterling could

improve the UK's current account balance. Identify one condition under which this improvement may not occur.

Question 6**[6 marks]**

Explain two reasons why the UK's reliance on "continued foreign portfolio inflows" (Extract 2.1) makes the economy vulnerable to a sudden loss of investor confidence.

Section Case Study 3. Indonesia — FDI, Resource Processing, and Exchange-Rate Management

Table 3.1: Selected Indonesia Macroeconomic Indicators, 2018–2023

Indicator	2018	2019	2020	2021	2022	2023
Real GDP Growth (%)	5.2	5.0	−2.1	3.7	5.3	5.0
CPI Inflation (%)	3.1	2.7	1.7	1.6	4.2	2.6
Current Account Balance (% GDP)	−2.9	−2.7	−0.4	0.3	1.0	−0.1
Nickel Exports (US\$ billion)	1.6	1.9	2.7	6.8	14.0	13.9
Net FDI Inflows (US\$ billion)	18.9	24.6	18.5	21.1	25.4	21.7
Bank Indonesia Policy Rate (% , year-end)	6.00	5.00	3.75	3.50	5.50	6.00
IDR/USD Exchange Rate (year-end)	14481	13901	14105	14269	15568	15397

Source: Adapted from Bank Indonesia, Indonesian Investment Coordinating Board (BKPM), and Indonesia Statistics Agency.

Extract 3.1: Indonesia's downstream gamble: nickel, FDI, and the rupiah

Indonesia banned the export of raw nickel ore in 2014, gradually tightening the restriction with a complete ban from January 2020. The policy, deeply controversial under World Trade Organization rules and the subject of an ongoing EU complaint, was designed to force foreign investors to build processing capacity inside Indonesia rather than ship raw ore overseas. The strategy has reshaped Indonesia's export structure: nickel exports rose from US\$1.6 billion in 2018 to US\$14.0 billion in 2022, with the bulk of growth coming from intermediate nickel products (ferronickel, nickel pig iron) and processed battery-grade nickel chemicals.

The downstream investment has been led primarily by Chinese firms. Tsingshan Holding Group, China's largest stainless-steel producer, has built the Morowali Industrial Park in Sulawesi, an integrated complex housing smelting, processing, and increasingly battery-precursor manufacturing. The total committed Chinese FDI in Indonesian nickel processing is estimated at over US\$30 billion. Net FDI inflows rose from an average of US\$20 billion before the pandemic to US\$25.4 billion in 2022, with nickel accounting for the bulk of the increase.

Yet the strategy has unresolved tensions. Bank Indonesia raised the policy rate from 3.50 per cent to 6.00 per cent between July 2022 and October 2023, in part to defend the rupiah from depreciation pressure. The rupiah weakened from 14269 to 15568 per US dollar in 2022, as the US Federal Reserve raised dollar rates faster than Bank Indonesia could match. "Currency stability is essential for FDI confidence, but defending the exchange rate raises borrowing costs for domestic firms," said Bank Indonesia Governor Perry Warjiyo. CPI inflation peaked at 5.95 per cent in September 2022, partly imported through the weaker rupiah, before being brought back into Bank Indonesia's 2–4 per cent target band by late 2023.

The longer-term question is whether the processing strategy will generate sustainable structural transformation. Critics argue that Chinese investors retain effective control over technology and management, that environmental costs from nickel smelting have been severe, and that the strategy leaves Indonesia exposed to a single demand source (the global electric vehicle battery supply chain). Supporters point to the share of nickel value-added now retained in Indonesia, which has tripled, and to the spillovers into supporting industries such as logistics, construction, and skilled-trade employment.

Source: Adapted from Reuters, 24 January 2024 and Indonesian Investment Coordinating Board Annual Report 2023.

Question 1**[2 marks]**

With reference to Table 3.1, calculate the percentage change in Indonesian nickel exports between 2018 and 2022. Show your working.

Question 2**[2 marks]**

Define "foreign direct investment (FDI)" and identify one piece of evidence from Extract 3.1 that the FDI is directed toward greenfield investment rather than financial portfolio flows.

Question 3**[4 marks]**

Using evidence from Extract 3.1 and Table 3.1, explain how the FDI-led downstream nickel strategy has affected Indonesia's current account balance.

Question 4**[4 marks]**

Explain how an increase in the Bank Indonesia policy rate from 3.50 per cent to 6.00 per cent could help to stabilise the rupiah against the US dollar.

Question 5**[6 marks]**

With the aid of an AD/AS diagram, explain how a sustained surge in FDI inflows of the magnitude experienced by Indonesia between 2020 and 2022 would affect both short-run and long-run aggregate supply.

Question 6**[6 marks]**

Explain two reasons why Indonesia's reliance on raw nickel export restrictions to attract FDI may not generate sustainable long-run economic development. Refer to Extract 3.1 in your answer.

Section Case Study 4. Vietnam — Export-Led Growth and the Inflation Pressure Test

Table 4.1: Selected Vietnam Macroeconomic Indicators, 2018–2023

Indicator	2018	2019	2020	2021	2022	2023
Real GDP Growth (%)	7.1	7.0	2.9	2.6	8.0	5.1
CPI Inflation, average (%)	3.5	2.8	3.2	1.8	3.2	3.3
Goods Exports (US\$ billion)	244	264	282	336	371	354
Goods Trade Balance (US\$ billion)	6.8	11.1	19.9	4.1	11.2	28.0
Net FDI Disbursements (US\$ billion)	19.1	20.4	19.9	19.7	22.4	23.2
State Bank of Vietnam Refinance Rate (%)	6.25	6.00	4.00	4.00	6.00	4.50
VND/USD Exchange Rate (year-end)	23200	23170	23115	22790	23615	24265

Source: Adapted from General Statistics Office of Vietnam, State Bank of Vietnam, and World Bank Vietnam Macro Monitoring 2023.

Extract 4.1: Vietnam's manufacturing miracle and the limits of catch-up growth

Vietnam has been one of the principal beneficiaries of the post-2018 “China plus one” supply-chain re-routing. Goods exports rose from US\$244 billion in 2018 to a peak of US\$371 billion in 2022, an increase of more than 50 per cent in four years, before slipping to US\$354 billion in 2023 as global demand weakened. Vietnam is now the second-largest exporter of mobile phones and a major exporter of textiles, footwear, and increasingly electronics components.

Foreign direct investment has been the principal vector. Samsung Electronics alone now operates the largest mobile phone manufacturing complex in the world in Bac Ninh and Thai Nguyen provinces, with cumulative investment of over US\$20 billion. Apple expanded contract production with Foxconn, Luxshare, and Goertek facilities, shifting an estimated 11 per cent of global iPhone production to Vietnam by 2023. The pattern is concentrated and FDI-intensive: roughly 70 per cent of Vietnamese goods exports originate from the foreign-invested sector, even though that sector employs only about a quarter of the workforce.

The State Bank of Vietnam has navigated this growth surge with a relatively conservative mone-

tary stance. The refinance rate was raised from 4.00 per cent to 6.00 per cent between September and October 2022 in response to global tightening and pressure on the dong, before being lowered to 4.50 per cent in 2023 as inflation moderated. CPI inflation peaked at 4.9 per cent on a year-on-year basis in early 2023 before settling within the government's target ceiling of 4 per cent.

Yet structural fragilities remain. The State Bank's flexibility is constrained by the heavy reliance on dollar-denominated trade and FDI flows. The dong has been managed within a tight band against the dollar, depreciating just 4 per cent in 2022 even as the yen and rupiah weakened by 20 per cent and 8 per cent respectively. The trade-off is that monetary policy has limited room to support domestic demand during downturns. A separate concern is the narrow base of value-added: critics argue that Vietnam's exports remain heavily reliant on imported intermediate inputs, particularly from China, and that domestic firms have limited integration into the foreign-invested supply chains. The government's recent strategic plan emphasises domestic-firm development and supplier-base deepening as priorities for the 2025–2030 period.

Source: Adapted from Vietnam Briefing, 8 January 2024 and World Bank East Asia and Pacific Economic Update, October 2023.

Question 1**[2 marks]**

With reference to Table 4.1, describe the trend in Vietnam's real GDP growth between 2018 and 2023.

Question 2**[2 marks]**

Define "net FDI disbursements" as used in Table 4.1.

Question 3**[4 marks]**

Using evidence from Extract 4.1, explain why FDI has been the principal driver of Vietnam's export growth.

Question 4**[4 marks]**

With reference to Table 4.1 and Extract 4.1, explain why Vietnam's CPI inflation rose from 1.8 per cent in 2021 to 3.2 per cent in 2022.

Question 5**[6 marks]**

With the aid of an AD/AS diagram, explain how sustained FDI-led export growth contributes to both short-run and long-run growth in Vietnam.

Question 6**[6 marks]**

Explain two reasons why Vietnam's high reliance on foreign-invested export sectors makes the economy vulnerable to external shocks. Refer to Extract 4.1 in your answer.

End of Booklet — Set B